Ex.No:1: Simulation of FDMA in MATLAB

Objective 1: To simulate an FDMA system by allocating individual frequency channels to multiple users and determine the channel bandwidth assigned to each user.
Output Parameter: Channel Bandwidth (kHz)

CODE:

```
clc;
clear;
close all;

fs = 1000;
t = 0:1/fs:1;

TBW = 600;
Users = 3;

CBW = TBW/Users;

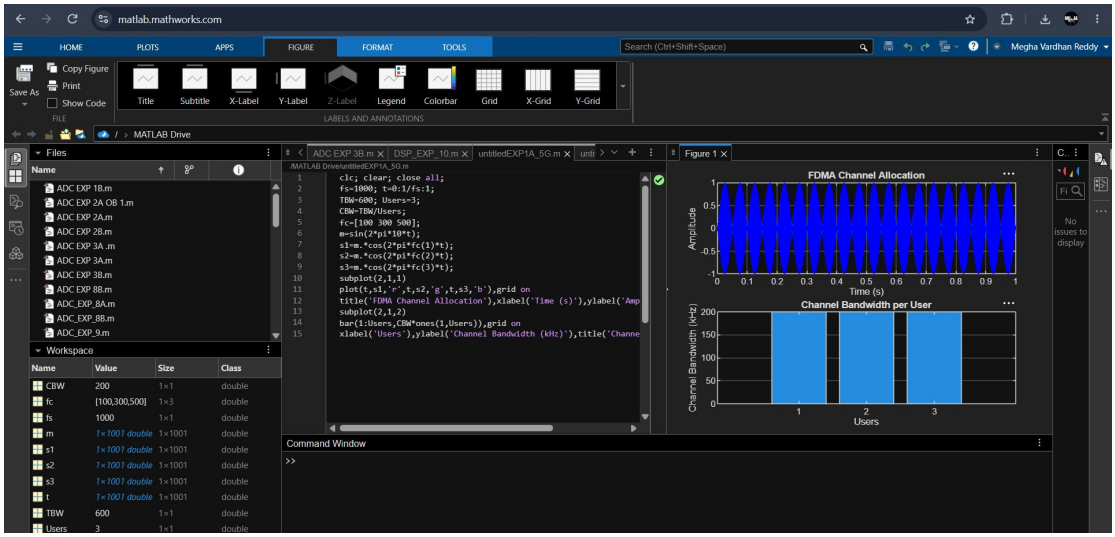
fc = [100 300 500];

m = sin(2*pi*10*t);

s1 = m.*cos(2*pi*fc(1)*t);
s2 = m.*cos(2*pi*fc(2)*t);
s3 = m.*cos(2*pi*fc(3)*t);

subplot(2,1,1)
plot(t,s1,'r',t,s2,'g',t,s3,'b');
grid on;
title('FDMA Channel Allocation');
xlabel('Time (s)');
ylabel('Amplitude');

subplot(2,1,2)
bar(1:Users,CBW*ones(1,Users));
grid on;
xlabel('Users');
ylabel('Channel Bandwidth (kHz)');
title('Channel Bandwidth per User');
```



Objective 2: To calculate the data rate achieved by each user in the FDMA system based on the allocated channel bandwidth. Output Parameter: Data Rate (kbps)

CODE:

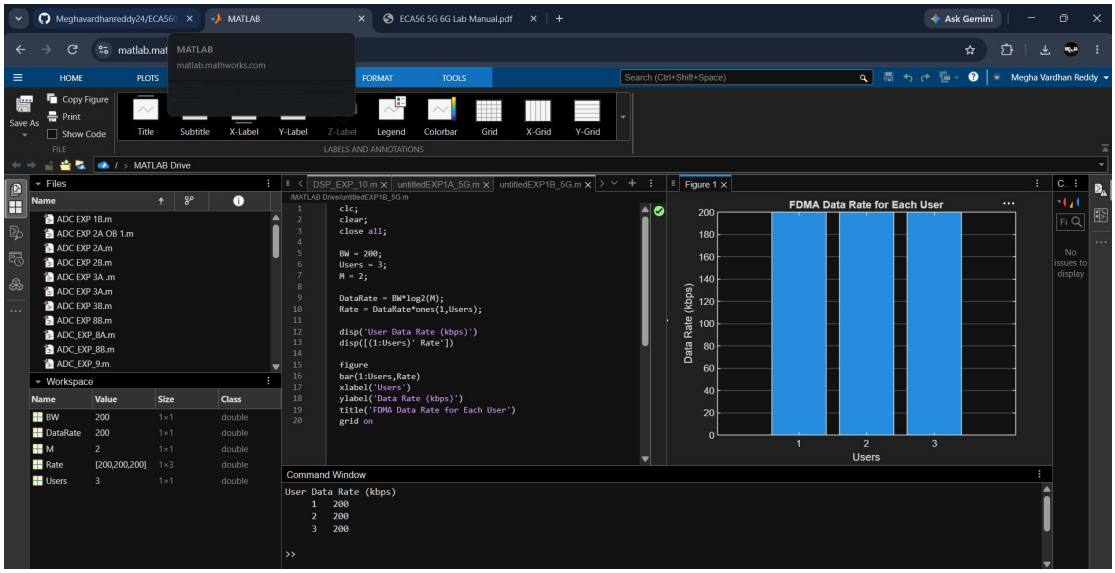
```
clc;
clear;
close all;

BW = 200;
Users = 3;
M = 2;

DataRate = BW*log2(M);
Rate = DataRate*ones(1,Users);

disp('User Data Rate (kbps)')
disp([(1:Users)' Rate'])

figure
bar(1:Users,Rate)
xlabel('Users')
ylabel('Data Rate (kbps)')
title('FDMA Data Rate for Each User')
grid on
```



Objective 3: To evaluate the communication quality of the FDMA system by calculating the Signal-toNoise Ratio (SNR) for different users. Output Parameter: Signal-to-Noise Ratio (SNR) (dB)

CODE

```
clc;
clear;
close all;

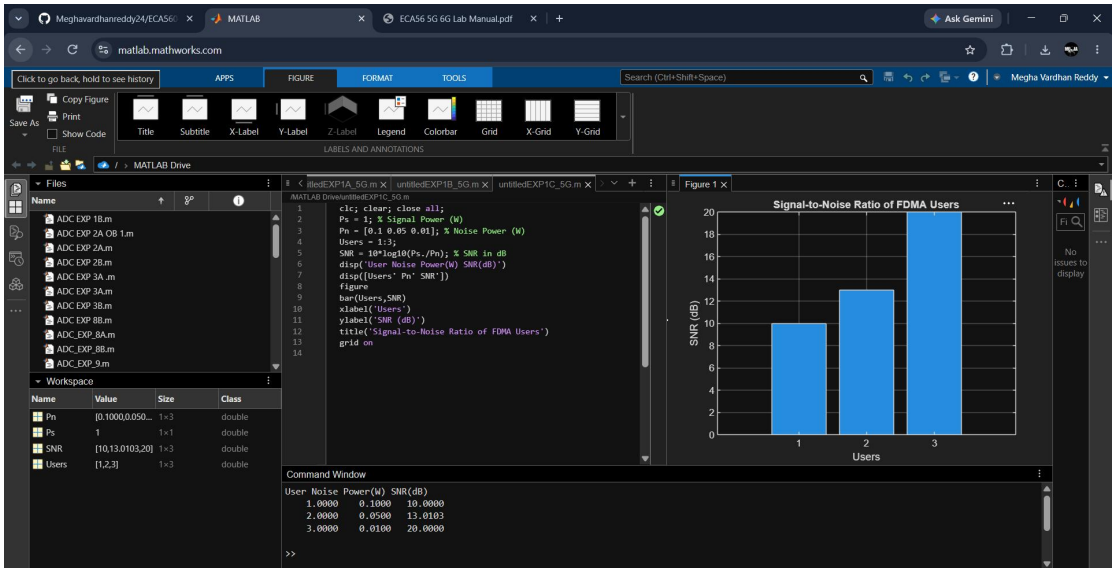
Ps = 1;
Pn = [0.1 0.05 0.01];

Users = 1:3;

SNR = 10*log10(Ps./Pn);

disp('User Noise Power(W) SNR(dB)')
disp([Users' Pn' SNR'])

figure
bar(Users,SNR)
xlabel('Users')
ylabel('SNR (dB)')
title('Signal-to-Noise Ratio of FDMA Users')
grid on
```


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Subtitle
X-Label
Y-Label
Z-Label
Legend
Colorbar
Grid
X-Grid
Y-Grid
LABELS AND ANNOTATIONS
MATLAB Drive
Files
Name
ADC EXP 1B.m
ADC EXP 2A OB 1.m
ADC EXP 2A.m
ADC EXP 2B.m
ADC EXP 3A .m
ADC EXP 3A.m
ADC EXP 3B.m
ADC EXP 8B.m
ADC_EXP_8A.m
ADC_EXP_8B.m
ADC_EXP_9.m
Workspace
Name
Value
Size
Class
Pn
[0.1000,0.050...
1×3
double
Ps
1
1×1
double
SNR
[10,13.0103,20]
1×3
double
Users
[1,2,3]
1×3
double
untitledEXP1A_5G.m
untitledEXP1B_5G.m
untitledEXP1C_5G.m
/MATLAB Drive/untitledEXP1C_5G.m
clc; clear; close all;
Ps = 1; % Signal Power (W)
Pn = [0.1 0.05 0.01]; % Noise Power (W)
Users = 1:3;
SNR = 10*log10(Ps./Pn); % SNR in dB
disp('User Noise Power(W) SNR(dB)')
disp([Users' Pn' SNR'])
figure
bar(Users,SNR)
xlabel('Users')
ylabel('SNR (dB)')
title('Signal-to-Noise Ratio of FDMA Users')
grid on
Figure 1
Signal-to-Noise Ratio of FDMA Users
SNR (dB)
Users
0
2
4
6
8
10
12
14
16
18
20
1
2
3
No issues to display
Command Window
User Noise Power(W) SNR(dB)
1.0000 0.1000 10.0000
2.0000 0.0500 13.0103
3.0000 0.0100 20.0000
>>